Lecture 19

Coadjoint orbits and the nilpotent orbit method

Objectives. *A representation acts on vectors. The coadjoint action of a Lie group instead acts on linear functionals on its Lie algebra. Each orbit of this action has a natural symplectic form. For connected simply connected nilpotent groups, these orbits label all irreducible strongly continuous unitary representations up to unitary equivalence. We prove the geometric construction and work through the Heisenberg example. The general classification is an explicit cited theorem.*

By the end of this lecture you will be able to:

- Compute a coadjoint action and its stabilisers.
- Construct the orbit symplectic form and check its signs.
- State the nilpotent orbit theorem with its global hypotheses.
- Choose a polarisation for a functional on a two-dimensional Heisenberg orbit and recover the Schrödinger representation by induction.

19.1 The dual action

Throughout, G is a finite-dimensional real Lie group with Lie algebra $\mathfrak{g}$. Its real dual is $\mathfrak{g}^* = \mathrm{Hom}_{\mathbb{R}}(\mathfrak{g}, \mathbb{R})$. We pair a functional ℓ and a vector X by $\ell(X)$. This dual is different from the complex dual of a representation space.

Definition 19.1. The *coadjoint action* of G on $\mathfrak{g}^*$ is

$$(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X).$$

The *coadjoint orbit* through ℓ is $O_\ell = \{g \cdot \ell : g \in G\}$. Its stabiliser is $G_\ell = \{g : g \cdot \ell = \ell\}$.

The inverse is needed for the action law. Indeed,

$$(g \cdot (h \cdot \ell))(X) = \ell(\mathrm{Ad}(h^{-1})\,\mathrm{Ad}(g^{-1})X) = \ell(\mathrm{Ad}((gh)^{-1})X) = ((gh) \cdot \ell)(X).$$

The stabiliser is closed by continuity. It is therefore an embedded Lie subgroup by the closed-subgroup theorem of Lecture 3. Differentiating the action gives

$$(\mathrm{ad}^*(X)\ell)(Y) = -\ell([X,Y]), \qquad \mathfrak{g}_\ell = \{X : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}. \tag{19.1}$$

The identity $\exp(tX)\cdot\ell = e^{t\,\mathrm{ad}^*(X)}\ell$ shows that $\exp(tX)\in G_\ell$ for every t exactly when $\mathrm{ad}^*(X)\ell = 0$. Thus $\mathrm{Lie}(G_\ell) = \mathfrak{g}_\ell$. For example, a central X lies in every $\mathfrak{g}_\ell$. Central directions therefore do not generate motion on an orbit.

We give $\mathcal{O}_\ell$ the smooth structure of G/G_ℓ. This uses the homogeneous-space theorem from Lecture 3. The natural map

$$G/G_\ell \longrightarrow \mathfrak{g}^*, \qquad gG_\ell \longmapsto g\cdot\ell$$

is injective and an immersion. At the identity coset, its derivative is induced by $X\mapsto \mathrm{ad}^*(X)\ell$, whose kernel is precisely $\mathfrak{g}_\ell$. At other points, equivariance gives the same rank. The orbit dimension is consequently $\dim\mathfrak{g} - \dim\mathfrak{g}_\ell$.

For a general Lie group, an orbit need not be an embedded submanifold of $\mathfrak{g}^*$. All tangent-space statements below use the quotient smooth structure. The Heisenberg orbits will be ordinary affine planes and points, so this distinction disappears there.

Write

$$\xi_X(\eta) = \left.\frac{d}{dt}\right|_0 \exp(tX)\cdot\eta = \mathrm{ad}^*(X)\eta, \qquad \eta\in\mathcal{O}_\ell.$$

The vectors $\xi_X(\eta)$ span the tangent space of $\mathcal{O}_\ell$ at η. Two choices of X give the same vector exactly when their difference belongs to $\mathfrak{g}_\eta$.

19.2 A symplectic form on each orbit

We need a small amount of differential-form language. A smooth 2-form ω assigns an alternating bilinear form ω_x to every tangent space, smoothly in local coordinates. On $\mathbb{R}^2$, for example,

$$(dp\wedge dq)((u_1,u_2),(v_1,v_2)) = u_1v_2 - u_2v_1.$$

A form ω is *nondegenerate* if, at every point x, the only tangent vector u at x with $\omega_x(u,v)=0$ for every tangent vector v at x is $u=0$.

In coordinates $x_1,\dots,x_d$, put $\omega_{ij} = \omega(\partial_i,\partial_j) = -\omega_{ji}$. Then $\omega = \frac{1}{2}\sum_{i,j}\omega_{ij}\,dx_i\wedge dx_j$. The form ω is *closed* when

$$\partial_i\omega_{jk} + \partial_j\omega_{ki} + \partial_k\omega_{ij} = 0 \quad \text{for all } i,j,k.$$

Equivalently, closedness means $d\omega = 0$. A closed nondegenerate 2-form is called a *symplectic form*. These are coordinate-independent conditions. We import this elementary differential-form calculus, including pullbacks and the exterior derivative d, from Warner, Chapter 2. We will state the identities used in the proof.

Theorem 19.2 (Orbit symplectic form)**.** *Every coadjoint orbit has a G-invariant symplectic form ω defined by*

$$\omega_\eta(\xi_X(\eta),\xi_Y(\eta)) = \eta([X,Y]). \tag{19.2}$$

This is the Kirillov–Kostant–Souriau form, with the sign convention of the infinitesimal action (19.1).

Proof. Well-definedness. If X changes by $A \in \mathfrak{g}_\eta$, then $\eta([A,Y]) = 0$. Changing Y has the same effect by antisymmetry. Thus the defining formula (19.2) depends only on the two tangent vectors. It is alternating and bilinear.

Nondegeneracy. If $\omega_\eta(\xi_X, \xi_Y) = 0$ for every Y, then $X \in \mathfrak{g}_\eta$ by the stabiliser formula (19.1). Hence $\xi_X(\eta) = 0$.

Smoothness. The fields ξ_X are smooth and span every tangent space. Choose a local frame from these fields. The defining formula (19.2) gives smooth coefficients in that frame, so ω is smooth.

Invariance. For $g \in G$, let g_* be the differential of the map $\eta \mapsto g \cdot \eta$ on the orbit. Since $g \exp(tX) = \exp(t\,\mathrm{Ad}(g)X)g$, differentiating $g \cdot (\exp(tX) \cdot \eta)$ at $t = 0$ gives

$$g_* \xi_X(\eta) = \xi_{\mathrm{Ad}(g)X}(g \cdot \eta).$$

Using that $\mathrm{Ad}(g)$ preserves brackets, we get

$$\begin{aligned} \omega_{g\cdot\eta}(g_*\xi_X, g_*\xi_Y) &= (g \cdot \eta)([\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y]) \\ &= \eta([X,Y]). \end{aligned}$$

So the form is invariant.

Closedness. Let $Q : G \to \mathcal{O}_\ell$ be the orbit map, given by $Q(g) = g \cdot \ell$. Put $\beta = Q^*\omega$. Invariance makes β a left-invariant form on G, and $\beta_e(X,Y) = \ell([X,Y])$. Exterior differentiation commutes with pullback. For a left-invariant 2-form, the differential at e is

$$(d\beta)_e(X,Y,Z) = -\beta_e([X,Y],Z) + \beta_e([X,Z],Y) - \beta_e([Y,Z],X).$$

We have now stated the two identities imported from Warner, Chapter 2. They are the commutation of d with pullback and the preceding formula for a left-invariant 2-form. Substitution gives

$$(d\beta)_e(X,Y,Z) = -\ell([[X,Y],Z]) + \ell([[X,Z],Y]) - \ell([[Y,Z],X]) = 0$$

by the Jacobi identity. Left invariance gives $d\beta = 0$ everywhere. Therefore $Q^*(d\omega) = 0$. Since Q is a submersion, every tuple of tangent vectors on the orbit lifts to G. This forces $d\omega = 0$. ∎

A nondegenerate alternating form exists only in even dimension. To see this, let A be the matrix of such a form in a basis. Then $A^T = -A$. If the size d of A were odd, then transposition invariance of the determinant would give $\det A = \det(-A) = -\det A$. Hence $\det A = 0$, and the form would be degenerate. In particular, every coadjoint orbit has even dimension. This is a useful first check on a proposed calculation.

Reading the form through linear functions

There is a useful way to remember the sign in the defining formula (19.2). For $X \in \mathfrak{g}$, restrict the linear function $\mu_X(\eta) = \eta(X)$ to an orbit. Its derivative in a tangent direction is

$$(d\mu_X)_\eta(\xi_Y) = (\mathrm{ad}^*(Y)\eta)(X) = -\eta([Y,X]) = \eta([X,Y]).$$

Thus $d\mu_X = \iota_{\xi_X}\omega$, where contraction means $(\iota_V\omega)(W) = \omega(V, W)$. This identifies the infinitesimal action using only the orbit form and the linear functions.

For a smooth real function f, nondegeneracy determines a unique tangent vector field X_f by $\iota_{X_f}\omega = df$. It is called the Hamiltonian vector field in this convention. The preceding calculation says $X_{\mu_X} = \xi_X$. Consequently the bracket

$$\{f, h\} = \omega(X_f, X_h)$$

satisfies $\{\mu_X, \mu_Y\} = \mu_{[X,Y]}$. We only need this identity for linear functions. It says that the Lie bracket is still visible after passing to an orbit.

A central element $W \in \mathfrak{g}$ has $\xi_W = 0$. Thus $d\mu_W = \iota_{\xi_W}\omega = 0$, so μ_W is locally constant on each orbit. It is constant when G is connected. Such a constant can still be the value of a bracket of linear functions. For the Heisenberg basis X, Y and Z of the next section, the identity above gives $\{\mu_X, \mu_Y\} = \mu_Z$, and μ_Z takes one constant value on each orbit.

A change of convention for Hamiltonian vector fields would change some of these signs. Here the definitions, rather than a remembered physics formula, determine them. This calculation concerns finite-dimensional smooth manifolds. It does not differentiate an unbounded operator or assert a quantisation rule.

19.3 Heisenberg planes and points

Let H be the Heisenberg group of Lecture 18, with Lie algebra $\mathfrak{h}$ spanned by X, Y and Z. We use exactly its matrix coordinates:

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab'), \qquad [X, Y] = Z.$$

In particular, c is a matrix entry. It is not the central coefficient in the logarithm, which is $c - ab/2$.

Write $\ell = (p, q, \lambda)$ when $\ell(X) = p$, $\ell(Y) = q$ and $\ell(Z) = \lambda$. The adjoint formulas of Lecture 18 are

$$\mathrm{Ad}(a, b, c)X = X - bZ, \qquad \mathrm{Ad}(a, b, c)Y = Y + aZ, \qquad \mathrm{Ad}(a, b, c)Z = Z.$$

The inverse of $g = (a, b, c)$ has first coordinates $(-a, -b)$. Hence $\mathrm{Ad}(g^{-1})X = X + bZ$ and $\mathrm{Ad}(g^{-1})Y = Y - aZ$. Applying ℓ to these vectors gives

$$(a, b, c) \cdot (p, q, \lambda) = (p + \lambda b, q - \lambda a, \lambda). \tag{19.3}$$

For $\lambda \neq 0$, this reaches every point of the plane with fixed last coordinate. Its stabiliser is the centre. For $\lambda = 0$, every point is fixed. Thus the orbit list consists of planes for nonzero λ and individual points $(p, q, 0)$.

On a plane, p and q are coordinates. Differentiating the action formula (19.3) gives

$$\xi_X = -\lambda\partial_q, \qquad \xi_Y = \lambda\partial_p, \qquad \xi_Z = 0.$$

The defining value of the form is $\omega(\xi_X, \xi_Y) = \ell(Z) = \lambda$. Since $(dp \wedge dq)(\xi_X, \xi_Y) = \lambda^2$, we obtain

$$\omega_\lambda = \frac{1}{\lambda}\, dp \wedge dq. \tag{19.4}$$

The coefficient is $1/\lambda$, not λ, because the vectors ξ_X and ξ_Y already carry a factor λ.

Point orbits have zero-dimensional tangent spaces. The zero form on such a tangent space is nondegenerate, since its kernel is the zero space. The orbit method will associate $(p, q, 0)$ with the one-dimensional character

$$\chi_{p,q}(a, b, c) = e^{i(pa+qb)}.$$

These are different representations for different (p, q). The trivial central character, with $\lambda = 0$, does not mean that the representation is trivial. Only $(0, 0, 0)$ gives the trivial character.

19.4 Polarisations and the orbit theorem

A Lie algebra is *nilpotent* if its lower central series terminates:

$$\mathfrak{g}_1 = \mathfrak{g}, \qquad \mathfrak{g}_{r+1} = [\mathfrak{g}, \mathfrak{g}_r], \qquad \mathfrak{g}_s = 0 \text{ for some } s.$$

For the Heisenberg algebra, $\mathfrak{g}_2 = \mathbb{R}Z$ and $\mathfrak{g}_3 = 0$. Nilpotence is stronger than solvability.

We import a global fact: for a connected simply connected nilpotent group, $\exp : \mathfrak{g} \to G$ is a diffeomorphism, and BCH is a finite polynomial. Every Lie subalgebra $\mathfrak{m}$ gives a closed subgroup $M = \exp \mathfrak{m}$. Indeed, finite BCH keeps products in $\mathfrak{m}$, and the exponential diffeomorphism takes the closed vector subspace $\mathfrak{m}$ to a closed subset. See Knapp, *Lie Groups Beyond an Introduction*, Section I.16.

Definition 19.3. A *real polarisation* for $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{m} \subset \mathfrak{g}$ such that

$$\ell([\mathfrak{m}, \mathfrak{m}]) = 0, \qquad \dim \mathfrak{m} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

A subspace $A \subseteq \mathfrak{g}$ is called *isotropic* for ℓ when

$$\ell([A, A]) = 0.$$

A real polarisation is thus an isotropic Lie subalgebra of the displayed dimension. This dimension says that the image of $\mathfrak{m}$ in $\mathfrak{g}/\mathfrak{g}_\ell$ is as large as the image of an isotropic subspace can be. In fact, the two displayed conditions force $\mathfrak{g}_\ell \subset \mathfrak{m}$, because an isotropic subspace of maximal dimension contains $\mathfrak{g}_\ell$. The subsection on the dimension of a polarisation below proves both statements.

For such a subalgebra, put

$$\chi_\ell(\exp W) = e^{i\ell(W)}, \qquad W \in \mathfrak{m}.$$

This is well-defined because exp is bijective. It is a character of M. Every BCH correction is a bracket of elements of $\mathfrak{m}$, so ℓ kills it and $\chi_\ell(\exp U \exp V) = \chi_\ell(\exp U)\chi_\ell(\exp V)$.

Theorem 19.4 (Nilpotent orbit theorem; cited)**.** *Let G be connected, simply connected and nilpotent. Every $\ell \in \mathfrak{g}^*$ has a real polarisation. The unitary induced representation*

$$\Pi_\ell = \operatorname{Ind}_{\exp \mathfrak{m}}^{G} \chi_\ell$$

is irreducible. Its equivalence class does not depend on the polarisation. Two such represen-

> *tations are equivalent exactly when their functionals lie on the same coadjoint orbit. Every irreducible strongly continuous unitary representation of G is unitarily equivalent to some Π_ℓ.*

This is Kirillov's theorem, imported from *Lectures on the Orbit Method*, Chapter 3. Induction means unitary induction with the quotient measure convention of Lecture 17. The statement concerns complex Hilbert representations, not just finite-dimensional modules.

The global assumptions matter. A central quotient can restrict which representations descend. A solvable group need not be nilpotent. Apart from the trivial group, no compact group and no semisimple group satisfies this theorem's hypotheses. We will meet related orbit constructions in Lecture 27, with a different integrality condition and a different representation model.

The dimension of a polarisation

We now isolate the linear algebra behind a polarisation. Define the alternating form

$$B_\ell(X, Y) = \ell([X, Y])$$

on $\mathfrak{g}$. Its kernel is $\mathfrak{g}_\ell$. Suppose $\dim O_\ell = 2d$ and $\dim \mathfrak{g}_\ell = k$. Then $\dim \mathfrak{g} = 2d + k$. The form B_ℓ induces a nondegenerate alternating form on the quotient $\mathfrak{g}/\mathfrak{g}_\ell$, which has dimension $2d$. A vector space with a nondegenerate alternating form is called a *symplectic vector space*.

Let L be a subspace of this quotient. Its *orthogonal* $L^\perp$ consists of the vectors whose induced pairing with every vector of L is zero. The subspace L is *isotropic* when the induced form vanishes on pairs of vectors of L, that is, when $L \subseteq L^\perp$. Nondegeneracy gives $\dim L^\perp = 2d - \dim L$, so an isotropic L has $\dim L \leq d$. Let $A \subset \mathfrak{g}$ be an isotropic subspace for ℓ. Its image in the quotient is isotropic, so this image has dimension at most d. Therefore

$$\dim A = \dim(A \cap \mathfrak{g}_\ell) + \dim\big((A + \mathfrak{g}_\ell)/\mathfrak{g}_\ell\big) \leq k + d.$$

Equality forces $\mathfrak{g}_\ell \subset A$. It also forces the image of A to be a d-dimensional isotropic subspace. The dimension in the definition of a polarisation is exactly $k + d$.

There is one extra demand: A must be a Lie subalgebra. A maximal isotropic vector subspace need not have this property. The existence of a suitable subalgebra is a substantial part of the nilpotent orbit theorem. The definition alone does not prove existence.

This count also predicts the number of variables in an induced model. If $\mathfrak{m}$ is a polarisation, then

$$\dim(G/M) = \dim \mathfrak{g} - \dim \mathfrak{m} = d = \tfrac{1}{2} \dim O_\ell.$$

The Heisenberg plane is two-dimensional, but its representation acts on functions of one real variable. The number of these variables is half the orbit dimension. The orbit does not single out which variables to use. A polarisation makes that choice.

19.5 Induction recovers the Schrödinger formula

Take $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Its stabiliser algebra is $\mathbb{R}Z$, so a polarisation must have dimension two. Choose

$$\mathfrak{m} = \operatorname{span}\{Y, Z\}, \qquad M = \{(0, b, c)\}, \qquad \chi_\ell(0, b, c) = e^{i\lambda c}.$$

The subalgebra $\mathfrak{m}$ is abelian and two-dimensional, so it satisfies both polarisation conditions. A section of $H/M \cong \mathbb{R}$ is $s(t) = (t, 0, 0)$. The quotient measure is dt, invariant under the induced translation action.

An induced function satisfies $f(gm) = \chi_\ell(m)^{-1} f(g)$. Write $\varphi(t) = f(s(t))$. Since

$$(a, b, c) = s(a)(0, b, c - ab),$$

its values are

$$f(a, b, c) = e^{-i\lambda(c-ab)}\varphi(a).$$

Left translation acts by $(\Psi_\lambda(g)f)(h) = f(g^{-1}h)$. For $g = (a, b, c)$,

$$g^{-1}s(t) = (t - a, -b, -c + ab).$$

Insert this into the preceding covariance formula. Its central expression becomes $-c + ab - (t - a)(-b) = -c + bt$. Therefore

$$(\Psi_\lambda(a, b, c)\varphi)(t) = e^{i\lambda(c-bt)}\varphi(t - a). \tag{19.5}$$

The operator is unitary: the phase has absolute value one and translation preserves dt. The general unitary induction construction of Lecture 17 supplies the representation law and strong continuity. Both can also be checked from this formula.

The sign of the translation differs from the Schrödinger formula of Lecture 18 because our section uses the left action on H/M. It gives an equivalent model. Let $R\varphi(x) = \varphi(-x)$. Then $R^{-1} = R$, and

$$(R\Psi_\lambda(a, b, c)R^{-1}u)(x) = e^{i\lambda(c+bx)}u(x + a) = \Pi_\lambda(a, b, c)u(x).$$

Thus the orbit theorem reproduces the Schrödinger representation with the precise convention already used. The plane's parameter λ determines its nonzero central character.

For a point $(p, q, 0)$, the polarisation is all of $\mathfrak{h}$. Induction from H to itself gives $\chi_{p,q}$. The plane and point calculations therefore recover the complete Heisenberg classification from Lecture 18. The general orbit theorem extends that pattern to every connected simply connected nilpotent group.

What the Heisenberg orbit list explains

The orbit calculation separates two phenomena. For $\lambda \neq 0$, changing p or q stays within one plane. It cannot change the equivalence class of the representation. For $\lambda = 0$, each point is an orbit of its own, so p and q both survive as representation labels. The discontinuous change in orbit dimension matches the change from infinite-dimensional representations to characters.

One can check the zero-charge conclusion without the general orbit theorem. In an irreducible unitary representation, every central group element acts by a scalar, by the unitary Schur lemma used in Lecture 18. If that scalar character is trivial, the representation factors through $H/Z(H) \cong \mathbb{R}^2$. An irreducible unitary representation of this abelian group is one-dimensional. Its continuous characters are $e^{i(pa+qb)}$, obtained by applying the one-parameter character classification to each coordinate.

For nonzero central character, the Stone–von Neumann theorem gives uniqueness directly. The orbit theorem recovers that result and explains its geometric label. It also shows what to try for a larger nilpotent group: calculate ad^*, find orbit invariants, determine the stabiliser dimension, choose a real polarisation for ℓ, and induce its character. The last exercise carries out the geometric steps in one four-dimensional example.

Self-checks

1. Why must the stabiliser algebra contain the centre?

Answer. If X is central, then $[X, Y] = 0$ for every Y. The stabiliser formula (19.1) puts X in $\mathfrak{g}_\ell$ for every ℓ.

2. Does the formula $\omega_\lambda = \lambda\, dp \wedge dq$ have the right normalisation?

Answer. No. Its value on $(-\lambda\partial_q, \lambda\partial_p)$ would be λ^3. The required value is λ, so the coefficient is $1/\lambda$.

3. Can one apply the orbit theorem directly to a Heisenberg group with circle centre?

Answer. No. That group is not simply connected. Apply the theorem to its simply connected cover. Then use the Descent theorem of Lecture 4. Only representations trivial on the covering kernel descend.

Summary

The coadjoint action turns $\mathfrak{g}^*$ into a union of even-dimensional symplectic orbits. For connected simply connected nilpotent groups, a polarisation converts an orbit into an irreducible induced representation. For the Heisenberg group, planes give Schrödinger representations and points give characters. The cited classification is exact in this setting. Its hypotheses are part of the result.

Reading

- Kirillov, *Lectures on the Orbit Method*, Chapters 2–3: Heisenberg representations, polarisations and the nilpotent correspondence.
- Knapp, *Lie Groups Beyond an Introduction*, Section I.16: simply connected nilpotent groups.

- Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Chapter 2: differential forms and exterior differentiation. The proof above uses only the two differential-form identities stated in that proof.